

Maintaining Draft Policies Under Meta Drift: What Retraining and Refresh Are Worth

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Abstract—Learned draft assistants deploy into games whose balance keeps changing. Using 1.95 million ranked Heroes of the Storm drafts spanning 45 game builds over 4.5 years, we measure what that drift costs a deployed draft policy and what each kind of maintenance buys back. In a replayed 2.8-year deployment, a model that is never maintained loses 1.06 percentage points of accuracy against one retrained after every build. Retraining every three builds recovers 1.01 of those points; recomputing the model’s aggregate statistics at every build, with the weights fixed, recovers 0.33. Once retraining is on a schedule, refreshing statistics between retrains adds only 0.02 to 0.04 points. Accuracy understates policy differences: two training regimes within 0.03 points of each other draft structurally broken teams at 17% and 32%. Head-to-head, scored on 298,000 games from five later builds that no agent’s statistics touch, a maintained agent’s picks average 0.34 percentage points more realized future-meta hero win rate than those of an agent trained at the same cutoff ($p = 0.07$ with value-function seed variance charged), and 1.28 points more than those of an agent trained two years earlier ($p < 10^{-4}$). The same-cutoff gap comes from how the value model was trained; three months of refreshed statistics add nothing measurable at the agent level. An outcome-shift detector fires on 16 of 21 build boundaries with documented changes and on 1 of 6 without (Fisher $p = 0.015$), yet as a refresh trigger it does no better than a calendar. Learned judges drift too: every judge trained before 2023 reverses the verdict of a 2026 contest.¹

Index Terms—Concept drift, dataset shift, MOBA drafting, model maintenance, non-stationarity.

I. INTRODUCTION

In prior work we built a draft assistant for Heroes of the Storm and deployed it in a browser where anyone can draft against it [1]. That paper froze the world on purpose: every experiment ran on a snapshot of 1.95 million ranked drafts, pinned in May 2026, so that results would be reproducible. The game did not hold still. Across the window our corpus covers, the developer shipped 45 game builds, and the balance changes among them moved the meta under every model we trained.

This paper restores the time axis and asks what a deployed draft policy loses when nobody maintains it, and what each maintenance action buys back. The two actions we price are retraining the model and refreshing the aggregate statistics its features are computed from (hero, pair, map, and composition win rates), with the weights left alone. Refresh is nearly free; retraining costs a training run. Practitioners would like to know whether the cheap action is enough.

We answer in four parts. First, a replayed 2.8-year deployment prices both actions at the accuracy level. Never maintaining costs 1.06 percentage points (pp) of accuracy

relative to retraining after every build. Retraining every three builds recovers 1.01pp of that, and refreshing statistics without retraining recovers 0.33pp. With retraining on a schedule, refresh between retrains adds 0.02 to 0.04pp, and every cadence from one to six builds lands within 0.11pp of retraining constantly. Second, accuracy understates what drift does to policies. Among eight training regimes, the patch-embedding regime comes within 0.03pp of our maintained model’s future accuracy, yet drafting greedily it assembles structurally broken teams twice as often (32% against 17%). Third, maintained and unmaintained MCTS agents draft against each other and are scored against realized outcomes from builds no agent has seen. Against an agent trained at the same cutoff, the maintained agent’s picks average +0.34pp of realized future-meta hero win rate ($p = 0.07$; +0.43pp, $p = 0.008$, with a single value function and more agent seeds). That gap comes from training the value model on statistics aligned with each game’s era, and three months of refreshed statistics add nothing measurable. Agents built with the same recipe one and two years earlier trail by 1.12 and 1.00pp; most of the loss arrives within a year. Fourth, the machinery a maintainer needs around these decisions: a detector of balance changes from outcomes alone, measured against the developer’s patch notes, and the observation that learned judges age like the models they judge.

Scoping notes. Each learned judge we tried scores this contest by its own era (Section VIII), so the head-to-head results rest on realized future outcomes and never on a learned judge. And the retrospective design can be tuned to the future it holds out, so we registered forward tests on OSF before submission (Section XI). Throughout, statistics used as training features are computed causally or out of fold, because statistics that include a row’s own outcome leak its label into its features (Section II).

II. SETTING, CORPUS, AND NAMES

We use the pinned corpus of our earlier work: 1,949,087 ranked Heroes of the Storm drafts on major patch 2.55, December 2021 through May 2026, a window over which the hero pool is fixed at 90. What is new is the time structure. The corpus spans **45 game builds**, of which 28 have at least 20,000 games (“sizable”). The balance unit is the build: hero tuning ships per build, and one marketing-level patch can span many builds. Build counts recur in this paper, nested as follows. Of the 45 builds, 28 are sizable. Forty builds map to scripted patch notes, and five have no published notes and are treated as no-change builds (Section VII). Consecutive sizable builds

¹<https://anonymous.4open.science/status/moba-draft-policies>

define 27 boundaries that are detectable in principle; the notes for 21 of them mention at least one hero and the notes for 6 mention none. Skill is bucketed into three tiers. Because of a labeling error in our replay collection, the snapshot’s tiers group Bronze and below (low), Silver, Gold, and Master (mid), and Platinum and Diamond (high), so the mid tier is not contiguous in league order. Every model here conditions on these labels, and the aggregate statistics are computed per label. Retraining the MAINTAINED value function with contiguous tiers (Bronze–Silver, Gold–Platinum, Diamond–Master; 47% of games change label) leaves future-window accuracy at 57.11% against 57.08% and the calibration slope at 0.93 against 0.93 (three seeds each), so no result depends on the grouping. Replacing the tier one-hot with continuous average match rating changes future accuracy by -0.017pp , and appending it changes it by $+0.007\text{pp}$, both inside seed noise.

All value functions share the architecture of our earlier work: a 283-input MLP over hero identities, map, tier, and aggregate-statistic features. The aggregate statistics come from our own corpus, per build and cumulatively to each build. The temporal split is a hard cutoff at build 2.55.14.95918 (February 2026). Training uses builds up to the cutoff; the *future window* is the seven strictly later builds, about 144,000 games. Where statistics come from matters in this paper, so we state it per arm. The MAINTAINED recipe gives every row statistics cumulative through the build before its own, which is causal by construction. The cutoff-frozen recipes give deployment rows the cumulative statistics at the cutoff, and their training rows the same statistics computed out of fold, as described next. Accuracies are labeled either *future-window* (the seven builds) or *deployment-weighted* (a 2.8-year replay, Section IV); the two windows give different numbers for the same policy.

Out-of-fold statistics. Giving every training row the statistics frozen at the cutoff leaks labels: a December 2021 game’s hero, pair, map, and composition win rates would then be computed from every game through February 2026, including that game, a known hazard with target statistics [8], [9]. Each replay therefore receives one of five hash folds, and a training row gets the cutoff statistics with its own fold’s games removed; deployment rows keep the unmodified cutoff statistics, which contain no deployment game. Built from the full counts, the same code reproduces the stored cutoff statistics to the last digit. The first version of this study used the leaky construction for every cutoff-frozen arm, and the correction changed results. For all-history training, in-period validation accuracy falls from 59.11% to 57.12%, the calibration slope on the future window (the slope of a logistic recalibration of the model’s predictions; 1 is calibrated) rises from 0.69 to 0.93, level with MAINTAINED’s 0.93, and the rate of structurally broken teams under greedy drafting falls from 43.5% to 25.8%. In-period validation cannot reveal the leak, because validation rows carry it too; the leaky models look calibrated there (slope 1.0). We report the leaky versions only as labeled controls, next to the leak-free results they replace.

The agent under maintenance has three learned parts. A win-probability model (the MLP above) scores completed drafts.

TABLE I
ARM NAMES. “CUTOFF” IS BUILD 2.55.14.95918 (FEBRUARY 2026).

Name	Meaning
MAINTAINED	value function trained on causal per-build statistics; its MCTS policy trained with kernel statistics through the last completed build (May 2026)
MAINTAINED-D90	as MAINTAINED, with statistics decayed on a 90-day half-life (Sec. IV)
MAINTAINED-AT-CUTOFF	the MAINTAINED value function, with its MCTS policy trained on kernel statistics frozen at the cutoff
UNMAINTAINED	value function trained on cutoff-frozen statistics with out-of-fold training rows; policy trained on cutoff statistics
STALE-1YR / STALE-2YR	the UNMAINTAINED recipe with the cutoff one and two years earlier, and an opponent model of that era
UNMAINTAINED-LEAKY	our first version of UNMAINTAINED, whose training rows saw their own outcomes; a control
LEAKY-FROZEN	the earlier paper’s model, whose statistics saw the future; a control

An opponent model, trained by behavioral cloning on human next-picks, supplies realistic adversaries. A policy network, trained by AlphaZero-style Monte Carlo tree search (MCTS) self-play against the opponent model with the win-probability model at the leaves, proposes picks and bans; the search kernel computes the win-probability features from a table of aggregate statistics. Evaluation uses two instruments. Head-to-head drafting: two agents alternate through the real 16-step ban-pick protocol, each acting greedily from its policy head, and the terminal drafts are scored against statistics of games actually played later (Section V). Structural metrics: the degenerate-composition rate counts teams missing a role that role-level statistics show is required (a healer or a frontline), and synergy and counter scores measure pairwise interactions beyond hero strength. These instruments come from our earlier work [1].

Because “frozen” can mean several things here, we name the arms once (Table I) and never use the word bare.

III. ACCURACY UNDERSTATES POLICY DAMAGE

Table II compares eight training regimes under identical architecture and budget: all-history training on cutoff-frozen statistics, recency windows of 3, 6, and 12 builds, exponential sample decay with 90- and 365-day half-lives, a learned patch embedding, and MAINTAINED, which trains on all history with statistics cumulative through each row’s previous build.

Leak-free, the regimes’ future-window accuracies span 56.48% to 57.08%, 0.6pp. Their rates of broken teams under greedy drafting span 16.8% to 49.7%, a factor of three. Accuracy says little about which regime is which. Across the seven cutoff-frozen regimes, the correlation between future accuracy and degeneracy is -0.46 ($n = 7$, not significant), and the patch-embedding regime, within 0.03pp of MAINTAINED’s accuracy, drafts broken teams at 31.9% against 16.8% (Welch

TABLE II

TRAINING REGIMES, LEAK-FREE, WITH THE LEAKY FIRST VERSION FOR COMPARISON (3 SEEDS EACH). VAL: IN-PERIOD VALIDATION ACCURACY; FUTURE: FUTURE-WINDOW ACCURACY; SLOPE: CALIBRATION SLOPE ON THE FUTURE WINDOW; DEGEN: BROKEN COMPOSITIONS UNDER GREEDY DRAFTING (3 SEEDS $\times$ 500 DRAFTS). MAINTAINED HAS NO LEAKY VERSION.

Regime	leak-free (out-of-fold)				leaky (first version)			
	Val	Future	Slope	Degen %	Val	Future	Degen %	
all history	57.1	56.59	0.93	25.8	59.1	56.21	43.5	
window 3	56.1	56.48	0.93	48.9	58.8	56.37	46.2	
window 6	56.9	56.63	0.94	46.7	59.0	56.53	49.7	
window 12	57.1	56.72	0.97	39.9	59.4	56.47	49.9	
decay 90d	56.5	56.66	1.03	49.7	58.9	56.48	50.1	
decay 365d	56.9	56.81	0.96	31.7	59.1	56.36	47.5	
patch embed	57.3	57.05	0.97	31.9	59.1	56.55	47.0	
MAINTAINED	57.2	57.08	0.93	16.8				

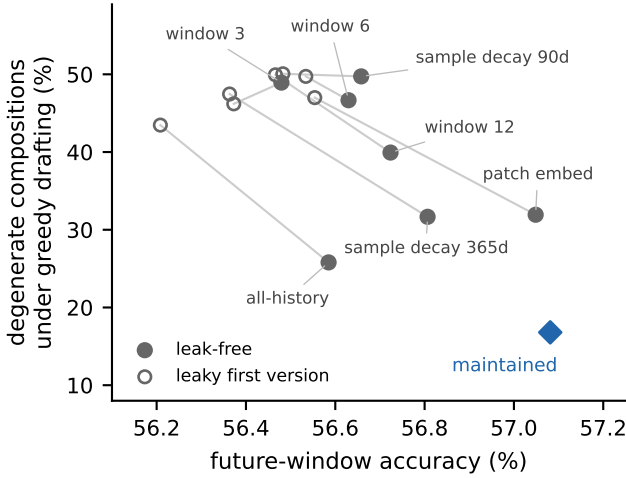


Fig. 1. Future-window accuracy against greedy degenerate-composition rate for the eight training regimes, leak-free (filled) and in the leaky first version (hollow, joined to their leak-free counterpart). Removing the leak moves the all-history regimes down and to the right; the short-history regimes stay degenerate.

$t = 10.9$, three seeds each). Selecting among these regimes by accuracy would call the best two interchangeable.

Two properties set the rate. The first is how much history a regime learns from: regimes that see mostly recent builds, by window or by 90-day sample decay, draft broken teams 40% to 50% of the time, and regimes that keep all history sit at 26% to 32%. The second is whether the training statistics match each game’s own era. Trained on all history with each row’s statistics cumulative through its previous build, MAINTAINED reaches 16.8%, against 25.8% for all-history training on cutoff-frozen statistics (Welch $t = 3.3$). Seed spread is large in some cells (standard deviation 4.7 points for all-history and 7.9 for the 12-build window), so we read only differences of this size. The opponent pool in this protocol is the earlier paper’s opponent model, which saw the test period; with an opponent model trained only to the cutoff, the leak-free all-history rate is 23.3% and MAINTAINED’s 14.1%.

Our first version reported a 2.5-fold gap (43.5% against 16.8%) and a correlation of +0.78 between accuracy and

TABLE III

DEPLOYMENT REPLAY OVER 2.8 YEARS: DEPLOYMENT-WEIGHTED REGRET AGAINST RETRAINING AFTER EVERY BUILD, ONE TRAINING SEED. “REFRESH” RECOMPUTES STATISTICS AT EVERY BUILD; “NO REFRESH” RECOMPUTES THEM ONLY AT RETRAINS. ^aTHREE SEEDS.

Policy	Retrain	Refresh	Regret (pp)
reference: retrain every build	35	35	0.000
never retrain, never refresh	0	0	+1.062
frozen-stats recipe, leak-free ^a	0	0	+1.05 to 1.12
frozen-stats recipe, leaky	0	0	+1.682
never retrain, refresh every build	0	35	+0.730
never retrain, refresh every 3rd	0	11	+0.738
never retrain, 14 even refreshes	0	14	+0.733
never retrain, detector refreshes	0	14	+0.743
retrain every 3, no refresh	11	11	+0.050
retrain every 3, refresh	11	35	+0.029
retrain every 6, no refresh	5	5	+0.106
retrain every 6, refresh	5	35	+0.067
finetune every 3, refresh	11	35	+0.089
retrain every 9 / 12, refresh	3 / 2	35	+0.20 / 0.22

degeneracy within the standard regimes, which we read as accuracy ranking policies backwards. The leak produced both (Figure 1). It raised the degeneracy of the all-history regimes most (from 25.8% to 43.5%, 31.7% to 47.5%, and 31.9% to 47.0%), left the short-history regimes near 47% to 50%, and lowered every regime’s future accuracy by 0.1 to 0.5pp. The claim that survives is weaker and still useful: at equal accuracy, policies can differ twofold in how often they draft a broken team, so policy metrics have to be measured directly.

IV. WHAT RETRAINING AND REFRESH ARE WORTH

We replay a 2.8-year deployment forward across 36 builds, starting from about 640,000 games of history at build 2.55.3.89754 (July 2023), and score every maintenance policy by games-weighted accuracy and by regret against retraining after every build (Table III). Every model in the replay trains on causal statistics with the same seed, so comparisons between rows use the same model family; differences under about 0.05pp are within the noise one training seed adds.

Doing nothing costs 1.06pp over the deployment. Refreshing the statistics every build, with the weights fixed, recovers 0.33pp of it, 31%. Retraining every three builds recovers 1.01pp even when the statistics are recomputed only at each retrain, and refreshing between retrains then adds 0.021pp; at a six-build cadence it adds 0.039pp. Retraining is the repair. Refresh is a cheap partial substitute when retraining is impossible, and nearly worthless once retraining happens on a schedule. Cadences from one to six builds land within 0.11pp of the reference; the regret grows past that (0.20pp at nine builds, 0.22pp at twelve). A warm-start finetune in place of a full retrain costs 0.06pp. In calendar terms the replay’s builds average 29 days apart, so a three-to-six-build cadence is quarterly to twice a year.

Refresh does little on the future window of the main split too. Holding the MAINTAINED checkpoints fixed and freezing their statistics at the cutoff lowers future-window accuracy by only 0.030pp (57.08% to 57.05%), a three-month horizon. The leak-free all-history model scores 56.59%. So at the 2026

cutoff, 0.46 of the 0.49pp that separates the two training recipes comes from training against statistics aligned with each game’s own era, and 0.03pp from refresh. In the replay the same recipe difference is worth about 0.02pp (1.08 against 1.06pp, mean of three seeds). The leaky construction had put the frozen-statistics recipe at 1.68pp, which our first version read as a large recipe effect. The recipe matters more at the later cutoff, where the model trains on four years of history instead of eighteen months, and cutoff-frozen statistics describe the early games worse.

Decayed aggregates. If recomputing statistics helps a little, old games may deserve less weight inside them. Decaying the aggregates with a 90-day half-life, with counts shrunk toward the cumulative value (100 pseudo-games), reaches 57.22% future-window accuracy against MAINTAINED’s 57.08%, and the unshrunk 90-day variant 57.19%. With 3 seeds per arm and the future games resampled, the gains are +0.14pp ($z = 2.5$, game-clustered) and +0.11pp ($z = 1.7$). The stronger evidence is a nested check that repeats the selection several builds earlier and scores a pseudo-future window: there the 90-day family wins by +0.19 and +0.21pp ($z = 3.3$ and 3.5). The variant chosen by that nested check is the unshrunk one (57.19%), so we recommend the family and report both. The gain is window-dependent in one respect: a 365-day half-life ties cumulative statistics on the future window (57.07%) and beats them by 0.14pp ($z = 3.1$) on the nested window. Under greedy drafting the two 90-day variants produce 18.9% and 19.1% broken teams against 16.8% for MAINTAINED, about 2 points higher, which three seeds per arm cannot resolve (Welch $t = 1.8$ and 2.3). Our decay is applied per build using each build’s last game date, so games within a build share one weight, and builds last up to 167 days.

Agents trained on the decayed value function draft better-timed heroes. In a direct head-to-head (fifteen seeds per side, 4,500 drafts) the MAINTAINED-D90 agent’s picks average +0.31pp of future-meta hero win rate over MAINTAINED’s ($z = 3.6$) on the out-of-sample builds of Section V, +0.15pp ($z = 1.9$) on 2.55.16.97039 and +0.44pp ($z = 4.1$) on the 2.55.17 builds, with no difference in broken teams ($z = 0.7$), synergy, or counters. Each of these two agents was trained on the single value-function seed with the best future-window accuracy of three, which favors the decayed agent by 0.19pp of accuracy at selection; the seeds within each arm differ by at most 0.14pp.

What refresh touches. A partial-refresh ablation on the fixed MAINTAINED checkpoints swaps statistics by signal class at test time. Every effect is within ± 0.07 pp over the three-month horizon. Refreshing hero-level statistics alone lowers accuracy by 0.07pp; refreshing pair and composition statistics alone raises it by 0.07pp, slightly more than refreshing everything (+0.03pp). We read this cautiously. Raw pair win rates drift as fast as hero win rates, and the pair features are normalized by hero win rates, so the pair-only cell also carries current hero strength, and the hero-only cell feeds the model mismatched numerators and denominators. Decaying each signal class at its own measured rate does not help either: hero-family statistics on a 90-day clock with pair and composition statistics on a 365-day clock score 57.07%, the

reverse assignment 57.15%, both below uniform 90-day decay.

V. THE COST OF NEGLECT, HEAD-TO-HEAD

Accuracy comparisons leave open whether any of this matters when policies play. We answer with direct competition. Two agents draft against each other in every pairing of their seeds, both first-pick orderings, 40 drafts per ordering over stratified maps and tiers, each agent acting greedily from its policy head. At play time no agent uses aggregate statistics; statistics enter only through the value function and the kernel’s statistics table during MCTS training. “Maintenance” at the agent level therefore means a policy trained against refreshed statistics, which is a retrain.

The agents share one MCTS recipe (200 simulations, 300,000 self-play episodes) and differ in their value function, the statistics table used during training, and the opponent model. MAINTAINED is our earlier agent, 15 MCTS seeds; its value function is the one of three MAINTAINED seeds with the best future-window accuracy, a selection worth at most 0.07pp of accuracy. A replication of MAINTAINED with 6 MCTS seeds, two on each of the three value-function seeds, charges value-function variance. UNMAINTAINED has 6 MCTS seeds, two on each of three leak-free value-function seeds, so its seed variance includes value-function variance. MAINTAINED-AT-CUTOFF holds MAINTAINED’s single value function and trains its policy on statistics frozen at the cutoff (5 seeds); it is the maintained recipe trained at the cutoff and then left alone. STALE-1YR and STALE-2YR use the leak-free value functions trained to their own cutoffs, statistics of that date, and an opponent model trained only on games up to that date, for both the policy’s initialization and its self-play opponent (3 and 6 seeds; the one-year arm was cut to one agent per value-function seed when compute was capped). The 2026 agents are all initialized from, and trained against, the earlier paper’s opponent model, which saw the test period; both sides of every 2026 matchup share it.

Each terminal draft is scored against the realized outcomes of later games: the average win rate of the heroes each side picked, the synergy of each side’s pairs and its counters against the other side (pair and matchup win rates beyond what the heroes’ own rates predict), and the rule-based degenerate-composition flag. The scoring games are the 298,428 ranked games of build 2.55.16.97039 and the four 2.55.17 builds (May through September 2026). No agent’s statistics touch them. Our first version of this protocol scored against all seven post-cutoff builds, six of which also fed MAINTAINED’s statistics; that overlap tripled the synergy gap. Inference uses crossed random effects for the two seed identities with a t reference and Satterthwaite degrees of freedom. The primary contrast is future-meta hero win rate; synergy, counters, degeneracy, and the other truth sets are secondary.

Against the leak-free unmaintained agent, the maintained agent’s 15 seeds average +0.43pp of future-meta hero win rate ($t = 3.3$, 10 df, $p = 0.008$; Table IV), about 2.3 points of team win probability once synergy and counters are included, and the advantage holds on both halves of the scoring set (+0.41pp on 2.55.16.97039, +0.46pp on the 2.55.17 builds). Those 15

TABLE IV

HEAD-TO-HEAD RESULTS ON OUT-OF-SAMPLE BUILDS. Δ : THE FIRST AGENT’S ADVANTAGE IN FUTURE-META HERO WIN RATE (PP, CROSSED SEED RANDOM EFFECTS, t REFERENCE). NET: TEAM WIN-PROBABILITY GAP IMPLIED BY HERO, SYNERGY, AND COUNTER SCORES TOGETHER, WITH WEIGHTS FITTED ON HELD-OUT REAL GAMES (MEAN OF TWO CROSS-FITTING SPLITS). M: MAINTAINED; M-D90: MAINTAINED-D90; M-CUT: MAINTAINED-AT-CUTOFF; U: UNMAINTAINED; S1, S2: STALE-1YR, STALE-2YR; M-VOL: MAINTAINED TRAINED ON S2’S ROW COUNT; U-LEAKY, S2-LEAKY: THE LEAKY FIRST VERSIONS.

Matchup	Seeds	Δ (pp)	t (df)	Net (pp)
M (all VF seeds) vs U	6 $\times$ 6	$+0.34 \pm 0.17$	2.0 (10)	+1.6
M vs U	15 $\times$ 6	$+0.43 \pm 0.13$	3.3 (10)	+2.3
M-d90 vs U	15 $\times$ 6	$+0.77 \pm 0.11$	7.0 (8)	+3.7
M vs M-cut	15 $\times$ 5	-0.05 ± 0.11	-0.5 (9)	-0.4
M-cut vs U	5 $\times$ 6	$+0.52 \pm 0.13$	3.9 (10)	+2.9
U vs S1	6 $\times$ 3	$+1.12 \pm 0.23$	4.8 (5)	+5.3
U vs S2	6 $\times$ 6	$+1.00 \pm 0.16$	6.3 (8)	+4.5
M vs S1	15 $\times$ 3	$+1.34 \pm 0.16$	8.6 (3)	+6.4
M (all VF seeds) vs S2	5 $\times$ 6	$+1.28 \pm 0.15$	8.7 (9)	+5.6
M vs S2	15 $\times$ 6	$+1.40 \pm 0.09$	15.9 (13)	+6.3
M-vol vs S2	5 $\times$ 6	$+1.28 \pm 0.16$	8.1 (7)	+5.9
<i>leaky controls (first version)</i>				
M vs U-leaky	15 $\times$ 15	$+0.69 \pm 0.10$	7.0 (32)	+2.6
M vs S2-leaky	5 $\times$ 5	$+1.10 \pm 0.17$	6.5 (8)	+4.0
U vs U-leaky	6 $\times$ 15	$+0.35 \pm 0.13$	2.8 (11)	+0.8

agents share one value function, so this estimate conditions on one value-function draw. The replication that varies the value-function seed gives $+0.34 \pm 0.17$ pp ($t = 2.0$, 10 df, $p = 0.07$). Its six agents range from -0.12 to $+0.82$ pp, and the 15 single-value-function agents from -0.01 to $+0.93$ pp, so agent-seed variance dominates at this effect size. We read the same-cutoff advantage as about a third of a point, positive in every estimate and significant only with the larger seed budget. Our first version measured $+0.66$ pp against the leaky agent ($+0.69$ pp at fifteen seeds per side), so about a third of the advantage we first reported came from the leak. The two unmaintained agents can be compared directly: the leak-free one beats the leaky one by $+0.35$ pp ($t = 2.8$) and drafts broken teams 28% of the time against 50%. The decayed-statistics agent leads the unmaintained agent by $+0.77$ pp ($t = 7.0$), consistent with its lead over MAINTAINED in Section IV.

The leak also produced two claims we retract. The first version found that unmaintained agents counter the maintained agent’s picks slightly better (-0.31 ± 0.08 in counter score at fifteen seeds). Against the leak-free agent the sign flips and the gap is small ($+0.12 \pm 0.07$ for MAINTAINED, $t = 1.7$), and the leak-free agent counters worse than the leaky one (-0.39 ± 0.09 , $t = -4.3$). The leaky value function leaned on its matchup features, which is where the leak concentrated. The first version also found that maintenance cut broken teams from 36.9% to 22.0% in head-to-head play. Against the leak-free agent the rates are 24.1% and 29.7% (-5.6 ± 7.6 points, not significant). At the agent level we cannot detect a degeneracy benefit from maintenance, although the greedy protocol of Table II shows one (16.8% against 25.8%). Synergy differences are small in every matchup ($+0.14 \pm 0.11$ here).

The MAINTAINED-AT-CUTOFF agent locates the advantage. It keeps MAINTAINED’s selected value function and trains its

policy on statistics frozen at the cutoff. Against MAINTAINED, whose policy trained on three more months of statistics, it is level (-0.05 ± 0.11 pp, $t = -0.5$). Against UNMAINTAINED, which differs from it only in how its value function’s training statistics were built, it leads by $+0.52 \pm 0.13$ pp ($t = 3.9$). The advantage comes from training the value function on statistics aligned with each game’s era, and three months of refreshed statistics add nothing we can detect, matching the 0.03pp that refresh adds on the accuracy side (Section IV). The first row of Table IV therefore compares training recipes: an agent trained once at the cutoff with the causal recipe already has the advantage. The cost of neglect proper is the gap between agents built with the same recipe at different cutoffs, which the staleness rows measure.

A year or two of staleness costs more, and most of it arrives in the first year. UNMAINTAINED, STALE-1YR, and STALE-2YR share the leak-free recipe and differ in their cutoff (February 2026, 2025, and 2024). UNMAINTAINED leads the one-year agent by $+1.12 \pm 0.23$ pp ($t = 4.8$, 5 df) and the two-year agent by $+1.00 \pm 0.16$ pp ($t = 6.3$), about 5 points of team win probability in both cases. Against MAINTAINED the one-year gap is $+1.34 \pm 0.16$ pp and the two-year gap $+1.40 \pm 0.09$ pp ($t = 15.9$), or $+1.28 \pm 0.15$ pp ($t = 8.7$) with value-function seeds varied. Because our corpus grows over time, the two-year agent also trained on fewer games; a maintained agent trained on STALE-2YR’s row count keeps $+1.28$ pp, 91% of the gap. Unlike our first version, the two-year agent shows a synergy deficit ($+0.21 \pm 0.08$ against MAINTAINED, $t = 2.8$; $+0.27 \pm 0.12$ against UNMAINTAINED; the one-year agent’s is similar, $+0.25 \pm 0.12$) and no counter advantage (-0.08 ± 0.10). The first version’s “synergy does not rot” and “the stale agent counters better” were properties of the leaky stale agent. One asymmetry remains: the 2026 agents were trained against the earlier paper’s opponent model, which saw the test period, while the stale agents’ opponent models stop at their own cutoffs, as a real stale deployment’s would. Some of the staleness gap may come from that difference; we did not run the 2026 agents with a cutoff-era opponent model. The leaky first version put the one-year gap at only $+0.56 \pm 0.21$ pp.

VI. WHAT DRIFTS

Per-build statistics let us measure how each signal class decays. Correcting each correlation for binomial sampling noise (disattenuated r), role-composition win rates are nearly static, at 0.95 with themselves over a lag of 16 builds. Hero pick rates are nearly as stable (0.89 at lag 16). Per-hero win rate is the fastest-decaying reliable signal, falling from 0.94 at lag 1 to 0.62 at lag 16 (raw correlations 0.85 and 0.57). Raw pair win rates track their component hero win rates (0.61 at lag 16). The pairwise interaction beyond the two heroes’ strengths is weak and mostly within-build sampling noise, a caution for pairwise features generally. The value model ages as well: its calibration slope on all games after its cutoff is 0.93 for the model trained at the 2026 cutoff, and 0.84 and 0.80 for models trained one and two years earlier (three seeds each).

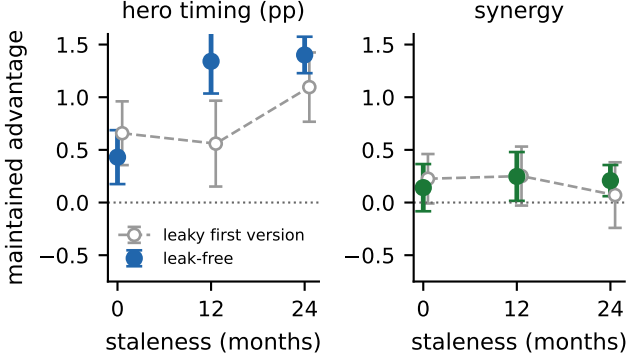


Fig. 2. Maintained agent’s advantage over unmaintained agents with the same cutoff (0 months) and cutoffs one and two years earlier, scored on out-of-sample builds (95% intervals, crossed seed random effects). Filled: leak-free agents. Hollow: the leaky first version.

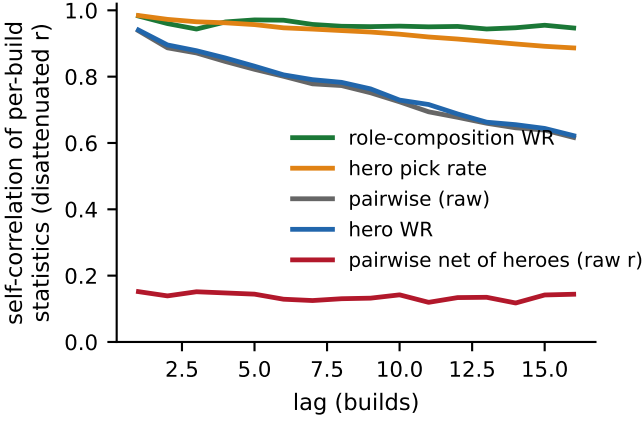


Fig. 3. Signal-class decay: self-correlation of per-build aggregate statistics against build lag, corrected for binomial sampling noise (disattenuated r), except the net pairwise series, which shows raw r because its reliability is near zero.

Two questions a maintainer asks after every patch have answers here. First, how soon can a new build’s own statistics be trusted? Measured against the build’s remaining games, which is what a deployed system can check, patch-local hero win rates come within a point of the truth after a median of about 17,200 games into a build. (A popular alternative, distance to the build’s final value, reaches the same threshold at 10,700 games, but the early games are part of the final value, and its error matches pure sampling error at every checkpoint, so it measures accumulation more than settling.) At our live inflow of roughly fifteen thousand ranked games per week, that is a little over a week. Second, does drift change which compositions never get played? The observed composition sets of consecutive builds have a mean Jaccard similarity of 0.81, and the per-build never-fielded count follows build volume ($r = -0.98$ against $\log$ games). Built from each era’s history, the never-fielded targets of our earlier paper’s synthetic-augmentation method stay within 13 compositions per tier of the full-corpus target from about three years of history on, and within 2 at the cutoff; the earliest eras differ

by up to 81.

VII. DETECTING BALANCE CHANGES FROM OUTCOMES

A maintainer wants to know when the game changed. The developer’s patch notes provide ground truth: we scripted 40 mapped patches, and five builds have no published notes and count as no-change builds. The detector tests each hero’s win rate across each boundary between sizable builds (two-proportion test, at least 200 games per side) and keeps the shifts that survive Benjamini–Hochberg correction at $q = 0.05$ within the boundary. A boundary fires when at least one hero survives.

The ground truth has two variants, and we report both. The scripted truth counts every hero the notes mention, including heroes mentioned only for bug fixes; this is what the frozen detection code uses. A balance-only variant drops bug-fix-only mentions, which appear in 37 of 59 patches (848 mentions). At the hero level, 46 of the detector’s 58 flags name a hero the notes mention (0.79), and 39 of 58 name a hero with a balance change (0.67). A randomly chosen hero at a random boundary is mentioned about 30% of the time. At the boundary level, the detector fires on 16 of the 21 boundaries whose notes mention a hero and on 1 of the 6 without (one-sided Fisher exact $p = 0.015$); under the balance-only truth, 15 of 19 against 2 of 8 ($p = 0.014$). Boundary precision, 16 of 17, sounds stronger than it is, because 21 of 27 boundaries carry mentions and random firing would already score 0.78 (binomial $p = 0.08$). The controls are few, and five of the six no-change boundaries fall inside one maintenance era (2.55.3). Two misses are roster-wide cleanup patches that mention 88 and 64 heroes; at 2.55.2.88122 all three flags fall outside the notes. Per-hero recall is low (0.06), because the notes list every individual change and most are too small to detect singly. The same test on pick and ban rates has hero-level recall 0.68 and precision 0.32; it picks up the community reacting, which spills across build boundaries and fires on hype as readily as on tuning.

How fast does it know? The first flagged hero at each firing boundary crosses an uncorrected $p < 0.05$ threshold a median of 2,500 games into the build. Over all 46 true-positive flags, the uncorrected first crossing has median 7,000 games. A Bonferroni-corrected sequential test never flags 7 of the 46 before the build ends, and flags the other 39 after a median of 18,900 games. The detector’s fire decisions themselves use the full build.

As a refresh trigger the detector is unremarkable. In the replay of Table III, refreshing a fixed model only when the detector fires (14 refreshes, each lagged by its first-detection latency) gives +0.743pp regret, against +0.733 for 14 evenly spaced blind refreshes, +0.738 for a refresh every third build, and +0.730 for every build. Latency itself costs 0.004pp; the sparser schedule costs 0.012pp. Recomputing statistics is cheap, so saving refreshes has little value. Sweeping the false-discovery level across two orders of magnitude moves hero-level precision from 0.90 (17 of 19 flags) to 0.45, and every operating point lands within 0.03pp of every other on the replay. Of the replay’s 36 builds, 20 are sizable; the detector fires on 14 of those 20 boundaries.

The detector’s value is as an alarm, and it needs little. It uses version identifiers, which almost every deployed system logs, and the magnitude of the outcome shift between versions; it needs no patch notes or labels. We also tested whether change timing could be recovered without version identifiers: a multi-scale changepoint detector over daily hero pick, ban, and win rates, tuned on pre-2025 boundaries (its best family was pick and ban rates), reaches held-out precision 0.18 at recall 0.27. With a ± 3 -day tolerance, the 11 held-out boundaries cover about 77 of 480 days, so chance precision is about 0.16. Without version identifiers we could not reconstruct the change calendar.

VIII. EVALUATORS DRIFT TOO

The obvious way to score the head-to-heads was the consensus of our win-probability models, the panel that adjudicated our earlier tournament. We did that first, got +0.8pp for the maintained side, and then noticed the number was unidentified: those judges were trained on a snapshot that includes the future period, so they share the maintained agent’s information advantage by construction. The fix produced a finding about judges.

We trained naive-feature judges (hero identities, map, and tier; no aggregate statistics) in two corpus families and scored one fixed set of 2,000 drafts, from our first head-to-head of MAINTAINED against UNMAINTAINED-LEAKY, with every judge. Here the drafts are only a test set; the question is how judges of different eras score the same drafts. Six ranked-play judges span early 2022 through the cutoff, each trained on the 314,761 most recent games before its cutoff (158,709 for the earliest, 2022Q1). These judges’ training data are subsets of both agents’ training data, and the 2026 judge’s window ends at the cutoff. Four Quick-Match judges (2021, 2022, 2024, 2026, on pools of 29K to 129K games) come from a mode with no draft phase: the matchmaker assembles teams, so these judges never saw a draft.

The ranked verdict rises with judge era: 0.469 (2022Q1) and 0.461 (2022-07), 0.493 (2023), 0.504 (2024), 0.495 (2025), and 0.500 (2026; 0.513 averaged over three seeds). The Quick-Match family repeats the pattern: 0.434 (December 2021) and 0.416 (2022), then 0.494 (2024) and 0.508 (2026). Clustering by the 25 seed pairings, standard errors are 0.005 to 0.013; judge seeds move a score by up to 0.011. Every judge trained before 2023 declares the unmaintained agent the winner of a 2026 contest, by 2.5 to 6 combined standard errors; from 2023 on, judges call it roughly even. The 2026 Quick-Match pool has no upper date bound, and 32% of its games postdate the cutoff; retrained with the pool capped at the cutoff, it scores 0.513 to 0.537 over three seeds, so the verdict does not change. What separates these two agents is largely current-meta knowledge, which an old judge lacks and mis-scores. Because the Quick-Match family moves from 0.416 to 0.508 with the game mode fixed, the reversal tracks the judge’s era within one corpus family.

The oldest and newest Quick-Match judges (December 2021 and 2026) both reproduce our earlier tournament’s ordering of 11 strategies at rank correlation 0.92. Contests decided by

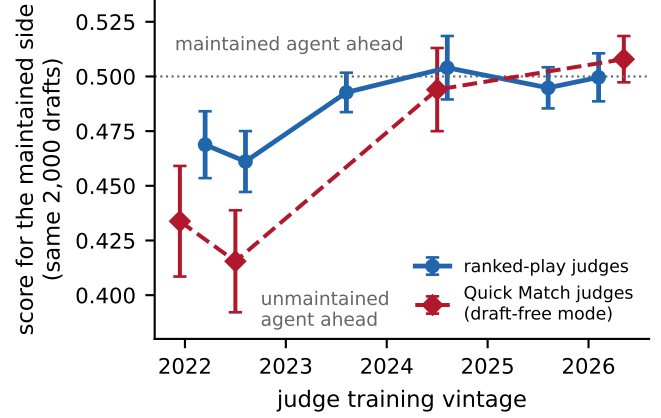


Fig. 4. The vintage matrix. Score assigned to the maintained side of the same 2,000 drafts by win-probability judges trained at different times (95% intervals clustered by seed pairing). Judges trained before 2023 reverse the verdict; later judges score the contest roughly even. Diamonds: judges trained on Quick Match, a mode without a draft phase.

TABLE V
OPPONENT MODELS SCORED ON 1.84M FUTURE NEXT-PICK SAMPLES (TOP-1/TOP-5 ACCURACY, %). ROWS 1 AND 4 SAW THE TEST DRAFTS IN TRAINING AND ARE REFERENCES ONLY. ^aRETRAINED BEFORE EACH TEST BUILD ON THE PRECEDING 30 DAYS OF PLAY.

Model (training data)	Top-1	Top-5
test-period fit (98% of the test drafts)	13.09	40.39
30 days before each build (rolling) ^a	11.97	–
30 days before the cutoff, trained once	11.83	36.98
earlier paper’s model (full snapshot)	11.71	36.31
last 6 builds before the cutoff	11.64	36.91
all history to the cutoff	11.08	34.47

composition structure are stable across eras, because structure barely drifts (Section VI); contests decided by meta timing are where judge vintage matters. For anyone evaluating agents in a drifting domain: date your judges, report them as a matrix, and put the headline on realized outcomes.

IX. THE OPPONENT MODEL DRIFTS ON ITS OWN CLOCK

A deployed drafter also carries a behavioral model of its opponent (the next-pick predictor that guides search), and player behavior drifts on a different schedule than balance. We compared opponent models on 1.84 million next-pick samples from the future builds (Table V).

With the same two model variants in each row, the earlier paper’s model scores 11.76% top-1, 0.68pp above the all-history model deployable at the cutoff; that is a lower bound on behavioral drift, since only about 7% of the earlier model’s training data postdates the cutoff. Recency recovers most of the gap: a model trained on the last six builds gains 0.56pp, and one trained once on the 30 days before the cutoff gains 0.75pp, finishing 0.08pp above the earlier paper’s model. The 30-day model’s lead is short-lived. It is the best model on the two 2.55.15 builds (13.1% and 12.6% top-1) and the worst on the 2.55.16 builds, falling to 10.6% by the last, below the all-history model there (11.2%). It wins on bans (17.1%

against 16.2%) and loses on picks. Rolled forward, retraining on the 30 days before each build, the window scores 11.97% overall, 0.14pp above the once-trained model, and 11.5% on the last build instead of 10.6%. It still trails the all-history model on one 2.55.16 build (10.7% against 11.0%), whose window held fewer games. Behavior wants a rolling window, retrained often; the all-history model is the safer fallback when recent play is thin.

Before running this comparison we set a rule for escalating it: retrain the MCTS pipeline on refreshed opponents if behavioral drift cost at least 1.0pp of top-1, or if swapping opponent pools moved the maintained value function’s greedy healer or degeneracy rates by at least 3pp. The rule lives in our analysis code, dated by file timestamps, and was not publicly registered. Neither condition was met: 0.63pp of top-1, and movements of 2.9 and 2.7pp (standard error about 1.9pp). The all-history value function, which the rule did not cover, moved more (5.8pp of healer rate, 3.5pp of degeneracy). By rule, the escalation did not run. For the two stale agents we did train era-matched opponent models, because an agent’s counter-play is learned from its opponent (Section V).

X. THE OPERATING POLICY

The measurements above reduce to a short policy. Train on causal or out-of-fold statistics and check calibration on rows the statistics never contained. Retrain the value model on a fixed schedule; every three to six builds (quarterly to twice yearly here) keeps regret within about 0.1pp of retraining constantly. Recompute the statistics at each retrain, and between retrains if it is free, which buys a few hundredths of a point. Prefer 90-day decayed statistics. Retrain the opponent model on recent play. Use the outcome-shift detector as an alarm that a build changed hero strength; do not use it to schedule refreshes.

Our production deployment retrains all three models monthly on the full corpus, with 90-day decayed statistics computed out of fold for training rows, and serves the same decayed statistics to the browser; a calibration-slope gate blocks deployment when the value model’s slope on held-out rows leaves [0.8, 1.25]. Builds have averaged about a month apart, so a monthly retrain is close to retraining after every build, more often than the replay says is necessary.²

XI. A PROSPECTIVE TEST, PRE-REGISTERED

Every result above is retrospective: the cutoff was chosen, then the future window was scored. So before submission we registered three forward claims on OSF (September 27, 2026; https://osf.io/3cxvd/overview?view_only=dbe817f56afe4507865e44bf24510748), with the analysis code frozen by commit hash: (1) the outcome-shift detector, decision rule untouched, fires on new builds whose patch notes contain hero changes and stays silent on no-change builds (builds whose changes are all individually minor are reported descriptively and adjudicate nothing); (2) the existing 4,500-draft head-to-head between MAINTAINED and

UNMAINTAINED-LEAKY, rescored with the out-of-sample protocol of Section V against the new builds’ realized statistics, keeps a positive maintained advantage in future-meta hero win rate (both agents and all drafts predate those builds, so nothing in this claim can be tuned); (3) every production refresh event is logged and reported against the detector’s fires. The claims cover only builds that ship after the registration date. The builds between our snapshot and the registration (2.55.16.97039 and the four 2.55.17 builds) already serve as the scoring set of Section V and are excluded. New ranked games arrive at about fifteen thousand per week, so a new build reaches the 20,000-game detector gate in under two weeks, and a six-month window should contain three or four sizable builds.

Two disclosures bear on the registered claims and will be repeated when they are reported. The registration text says bug-fix-only and ARAM-only hero mentions are excluded from the ground truth, as in the retrospective analysis. The frozen code does not exclude them: it counts every hero the notes mention. The registered test is the frozen code, so C1 will be adjudicated with the any-mention truth, and we will also report the balance-only variant. The retrospective numbers in Section VII report both. Second, the registered head-to-head (C2) was drafted by UNMAINTAINED-LEAKY, the leaky first version of the unmaintained agent, because it was registered before the leak was found. We will report it as registered and report the leak-free matchups of Section V alongside it as descriptive results.

XII. RELATED WORK

Concept drift and dataset shift have a large literature, most of it classifier-centric: detect distributional change, adapt the model, measure accuracy [2], [3], [4]. The canonical detectors monitor a supervised error stream (DDM tracks error-rate inflation [6]; ADWIN maintains adaptive windows over it [7]), and adaptation means retraining or reweighting the classifier. Change-point detection supplies the standard machinery [5]; our detector is deliberately plain, and what we add is the ground truth we score it against (developer patch notes) and its price as a maintenance trigger. Our results sit on the retraining side of that literature: in this domain, retraining on a schedule does almost all of the repair, and the policy-level cost of drift is larger than the accuracy cost suggests. Target leakage through aggregate statistics is a known failure in tabular learning [8], and ordered or out-of-fold target statistics are the standard remedy [9]; we show how it distorts a drift study specifically, by making the unmaintained baseline look worse than drift does.

Continual and non-stationary reinforcement learning study policies that keep learning as the environment changes [10], [11], with catastrophic forgetting as a central obstacle [12]. The deployed case we study fixes the policy between maintenance events and asks when and what to touch; a continual learner would still need the trigger, the evaluator hygiene, and the leak-free statistics this paper prices. OpenAI Five kept one Dota 2 agent current across game patches by “surgery” on its network and continued training [13], a retraining-side answer to the same problem at far larger scale. On the

²<https://draft-assistant-review.vercel.app>

operations side, data dependencies and feature freshness are a known maintenance surface of deployed ML [14], production-readiness rubrics score monitoring loops [15], data-validation systems treat stale statistics as a failure [16], and practitioners report retraining cadence as an open question [17]; we supply a measured decomposition of what refresh and retraining each buy in one domain.

In games, drafting has been treated as prediction and recommendation, from pick prediction in Dota 2 [18] to team-aware hero recommendation [19] and personalized draft recommendation [20]. In this journal, Chen et al. learn to draft with neural networks and MCTS [21], and Gourdeau and Archambault select heroes in professional Heroes of the Storm and Dota 2 drafts with a discriminative network [22]. Balance evaluation has its own methods [23]. To our knowledge none of this work measures maintenance policies against patch-note ground truth or prices staleness in head-to-head play. Offline reinforcement learning under non-stationarity is usually posed as a training-time problem [24]; here the policy is fixed and the world moves. This paper is the time-axis companion to our draft-policy paper [1]. Player-skill drift is out of scope and is the subject of a companion study of personalization.

XIII. LIMITATIONS

Everything here is one game and one major patch, a scoping that buys 45 builds with a fixed hero pool at the cost of external validity. The deployment replay uses one training seed per cell; orderings among regrets within about 0.05pp (for example 0.029, 0.050, and 0.067) are at noise level, although the paired comparisons on one model, such as the detector against blind schedules, are not. The judge-free protocol rests on realized hero, pair, and matchup statistics, which are marginal quantities; we mitigate by triangulating the metrics and by combining them with weights fitted on held-out real games. Agent-level comparisons carry three caveats. The 2026 agents share the earlier paper’s opponent model, which saw the test period, while the two-year agent’s opponent model is era-matched, so part of the two-year gap may come from the opponent model. MAINTAINED’s value function was selected on the future window from three seeds, and MAINTAINED-AT-CUTOFF shares it, so the split between them conditions on one value-function draw; with value-function seeds varied, the same-cutoff advantage is not significant ($p = 0.07$). The leak-free staleness gradient’s one-year point rests on three agent seeds. We tested about twenty accuracy cells on the same future window and many matchup-by-metric combinations; we name the primary contrasts in each section and treat the rest as secondary. Finally, our analysis went wrong three times before this version: an evaluator panel that shared the maintained agent’s vintage (Section VIII), scoring games that overlapped the maintained agent’s statistics (Section V), and unmaintained baselines trained on their own outcomes (Section II). The last one changed the paper’s main conclusion, and we cannot rule out further errors of the same kind.

XIV. CONCLUSION

A draft policy that nobody maintains loses about a point of accuracy over three years of balance changes. Retraining

on a schedule of three to six builds recovers nearly all of it; refreshing statistics with the weights fixed recovers a third, and adds a few hundredths of a point once retraining happens. At the agent level, three months without refresh cost nothing measurable, and one or two years cost about a point of future-meta hero win rate. Accuracy understates the policy-level damage, and learned judges are as era-bound as the models they score.

AI DISCLOSURE

Portions of this manuscript’s text were drafted and edited with Claude (Anthropic) [25]; all experiments, analyses, and claims were designed, verified, and approved by the authors.

REFERENCES

- [1] Anonymous authors, “Diagnosing and repairing out-of-distribution failure in MOBA draft policies,” under review, 2026.
- [2] J. Quiñero-Candela, M. Sugiyama, A. Schwaighofer, and N. D. Lawrence, Eds., *Dataset Shift in Machine Learning*. MIT Press, 2009.
- [3] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, “A survey on concept drift adaptation,” *ACM Computing Surveys*, vol. 46, no. 4, 2014.
- [4] J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, and G. Zhang, “Learning under concept drift: A review,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 31, no. 12, pp. 2346–2363, 2019.
- [5] S. Aminikhanghahi and D. J. Cook, “A survey of methods for time series change point detection,” *Knowledge and Information Systems*, vol. 51, no. 2, 2017.
- [6] J. Gama, P. Medas, G. Castillo, and P. Rodrigues, “Learning with drift detection,” in *Brazilian Symposium on Artificial Intelligence (SBIA)*, 2004.
- [7] A. Bifet and R. Gavaldà, “Learning from time-changing data with adaptive windowing,” in *Proc. SIAM International Conference on Data Mining*, 2007.
- [8] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, “Leakage in data mining: Formulation, detection, and avoidance,” *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, 2012.
- [9] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, “CatBoost: Unbiased boosting with categorical features,” in *Advances in Neural Information Processing Systems*, 2018.
- [10] S. Padakandla, “A survey of reinforcement learning algorithms for dynamically varying environments,” *ACM Computing Surveys*, vol. 54, no. 6, 2021.
- [11] K. Khetarpal, M. Riemer, I. Rish, and D. Precup, “Towards continual reinforcement learning: A review and perspectives,” *Journal of Artificial Intelligence Research*, vol. 75, pp. 1401–1476, 2022.
- [12] J. Kirkpatrick et al., “Overcoming catastrophic forgetting in neural networks,” *Proceedings of the National Academy of Sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.
- [13] C. Berner et al., “Dota 2 with large scale deep reinforcement learning,” arXiv:1912.06680, 2019.
- [14] D. Sculley, G. Holt, D. Golovin, E. Davydov, T. Phillips, D. Ebner, V. Chaudhary, M. Young, J.-F. Crespo, and D. Dennison, “Hidden technical debt in machine learning systems,” in *Advances in Neural Information Processing Systems*, 2015.
- [15] E. Breck, S. Cai, E. Nielsen, M. Salib, and D. Sculley, “The ML test score: A rubric for ML production readiness and technical debt reduction,” in *IEEE International Conference on Big Data*, 2017.
- [16] N. Polyzotis, M. Zinkevich, S. Roy, E. Breck, and S. Whang, “Data validation for machine learning,” in *Proc. Machine Learning and Systems (MLSys)*, 2019.
- [17] S. Shankar, R. Garcia, J. M. Hellerstein, and A. G. Parameswaran, “Operationalizing machine learning: An interview study,” arXiv:2209.09125, 2022.
- [18] A. Summerville, M. Cook, and B. Steenhuisen, “Draft-analysis of the Ancients: Predicting draft picks in DotA 2 using machine learning,” in *AIIDE Workshop on Experimental AI in Games*, 2016.
- [19] Z. Chen, T.-H. D. Nguyen, Y. Xu, C. Amato, S. Cooper, Y. Sun, and M. S. El-Nasr, “The art of drafting: A team-oriented hero recommendation system for multiplayer online battle arena games,” in *Proc. ACM Conference on Recommender Systems (RecSys)*, 2018.

- [20] H. Lee, D. Hwang, H. Kim, B. Lee, and J. Choo, “DraftRec: Personalized draft recommendation for winning in multi-player online battle arena games,” in *Proc. ACM Web Conference (WWW)*, 2022.
- [21] S. Chen, M. Zhu, D. Ye, W. Zhang, Q. Fu, and W. Yang, “Which heroes to pick? Learning to draft in MOBA games with neural networks and tree search,” *IEEE Transactions on Games*, vol. 13, no. 4, pp. 410–421, 2021.
- [22] D. Gourdeau and L. Archambault, “Discriminative neural network for hero selection in professional Heroes of the Storm and DOTA 2,” *IEEE Transactions on Games*, vol. 13, no. 4, pp. 380–387, 2021.
- [23] A. Jaffe, A. Miller, E. Andersen, Y.-E. Liu, A. Karlin, and Z. Popović, “Evaluating competitive game balance with restricted play,” in *Proc. AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, 2012.
- [24] S. Levine, A. Kumar, G. Tucker, and J. Fu, “Offline reinforcement learning: Tutorial, review, and perspectives on open problems,” [arXiv:2005.01643](https://arxiv.org/abs/2005.01643), 2020.
- [25] Anthropic, “Claude (Fable 5),” large language model, 2026.